

Appendix CM-v20: Core Micro-Theorem Pack

Canonical v20 Pack: Finite Detector Certificates, Higher Ext Edges, Transient Defects, and Reduction Closure

Version CM-v20.0.0

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Normative status. This appendix is subordinate to AK-HPDST v20.0.0 Main Chapters 1–8, the corrected Foundation Appendices A–G, and the CR-v20 baseline register. It extracts small reusable Core theorems at exactly their proved or displayed conditional strength. It does not enlarge the Core, create an external-domain theorem, infer a Return Theorem, repair missing evidence numerically, or promote a detector observation, synthetic duality, Search artifact, specification, operational rule, or companion template to theorem evidence.

Conservative-extension rule. CM-v20 preserves the repaired v19 metric, crop, representative, actual-morphism Type IV, Restart, and governance theorems without weakening their hypotheses. The v20 extension adds repaired-zero transport, finite detector soundness and completeness, family-relative proof compression and no-go bounds, canonical higher field-coefficient Ext edges, conditional higher discharge and natural zero-reflection, transient-defect detection, and exact finite-to-infinite reduction. Soundness is never read as completeness; terminal Type IV zero is never read as full defect vanishing; and a finite prefix is never read as an infinite reduction certificate.

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1 Scope, dependency baseline, and claim discipline

Policy 1.1 (CM-v20-001: Core micro-theorem boundary and no external upgrade). A CM-v20 item may be consumed only with its CM UID, declared role, evidence status, comparison policy, complete hypotheses, source Claim UIDs, artifact identities, and failure routes. A CM-v20 theorem is internal to the declared v20 Core regime and yields no external-domain conclusion without a separately proved and registered Interface or Bridge Theorem.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-001
Role / status: operational_policy / operational
Source claims: CR-G-001, CR-G-002, CR-G-017, CR-G-018
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-001; inherited at immutable v19 strength.

Definition 1.2 (Standing repaired Core regime). Fix a field k . Work in the constructible one-parameter regime $\text{Pers}_k^{\text{cons}}$. For a finite-type filtered object F , monitored degree i , right-open window W , and threshold $\tau > 0$, define

$$B_{i,W}(F) := W_W P_i(F), \quad \text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(B_{i,W}(F)).$$

The standard bottleneck metric with the inherited repaired finite/infinite-bar convention is used whenever metric comparison is invoked. Every detector record additionally declares one source stage among

windowed_prethreshold, repaired_postthreshold, kernel_defect, cokernel_defect.

A theorem proved at one stage is not silently transported to another stage.

Definition 1.3 (Typed CM status). Every CM item records a document role, evidence status, invocation role, conformance status, claim-level comparison policy, admissible runtime modes, source binding, and structured failure routes. Missing evidence gives undefined; it is never silently encoded as a numbered obstruction, zero, pass, not_invoked, or not_compared.

2 Barcode algebra and categorical guard-rails

Lemma 2.1 (CM-v20-002: Infinite-bar threshold invariance). *For every finite $\tau > 0$,*

$$T_\tau^{\text{bar}}([a, \infty)) = [a, \infty).$$

An infinite bar never lies in a finite threshold band $[\tau - \varepsilon, \tau + \varepsilon]$.

Proof. Its threshold length is $+\infty$, while hard deletion removes only finite bars of length at most τ . □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-002
Role / status: core_lemma / proved
Source claims: CR-A-005, CR-A-010; CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-002; inherited at immutable v19 strength.

Theorem 2.2 (CM-v20-003: Hard threshold deletion idempotence). *For every barcode profile B and $\tau > 0$,*

$$T_\tau^{\text{bar}} T_\tau^{\text{bar}}(B) = T_\tau^{\text{bar}}(B).$$

Proof. After the first deletion no finite bar of length at most τ remains. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-003
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-009, CR-A-010
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-003; inherited at immutable v19 strength.

Theorem 2.3 (CM-v20-004: Hard threshold composition). *For $\tau, \sigma > 0$,*

$$T_{\sigma}^{\text{bar}} T_{\tau}^{\text{bar}}(B) = T_{\max\{\tau, \sigma\}}^{\text{bar}}(B).$$

Proof. The composite deletes exactly the finite bars whose length is at most τ or at most σ , equivalently at most $\max\{\tau, \sigma\}$. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-004
Role / status: core_micro_theorem / proved
Source claims: CR-CH1-018, CR-A-010; CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-004; inherited at immutable v19 strength.

Theorem 2.4 (CM-v20-005: Retained-profile threshold monotonicity). *If $0 < \tau \leq \sigma$, every bar retained at threshold σ is retained at threshold τ .*

Proof. A finite length greater than σ is greater than τ ; infinite bars survive both thresholds. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-005
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-009; CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-005; inherited at immutable v19 strength.

Theorem 2.5 (CM-v20-006: Positive-threshold short bars are not Serre). *For $\tau > 0$, the class of constructible persistence modules whose bars all have finite length at most τ is not extension-closed in general. Hence hard positive-threshold deletion is not globally licensed as a Serre localization or exact reflector.*

Proof. At $\tau = 1$, the exact sequence

$$0 \rightarrow k_{[1,2)} \rightarrow k_{[0,2)} \rightarrow k_{[0,1)} \rightarrow 0$$

has short subobject and quotient but a length-two middle interval. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-006
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-006, CR-A-008, CR-A-009
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-006; inherited at immutable v19 strength.

3 Window restriction, crop order, and repaired evaluation

Theorem 3.1 (CM-v20-007: Nested window restriction identity). *If $W' \subseteq W$, then at barcode-profile level*

$$W_{W'}W_W(B) = W_{W'}(B).$$

In particular, W_W is idempotent.

Proof. For each bar I , $(I \cap W) \cap W' = I \cap W'$. Empty intersections and multiplicities agree. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-007
Role / status: core_micro_theorem / proved
Source claims: CR-CH1-022, CR-A-031; CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-007; inherited at immutable v19 strength.

Policy 3.2 (CM-v20-008: Fixed window-first repaired evaluation). The canonical evaluation order is

$$F \mapsto P_i(F) \mapsto W_W P_i(F) \mapsto T_\tau^{\text{bar}}(W_W P_i(F)).$$

Threshold clearance is measured after windowing and before deletion. A crop-after-collapse expression is not substituted unless CM-v20-010 is discharged.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-008
Role / status: operational_policy / operational
Source claims: CR-CH1-023, CR-CH2-033, CR-A-037
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-008; inherited at immutable v19 strength.

Theorem 3.3 (CM-v20-009: No unconditional crop-collapse commutation). *In general,*

$$W_W T_\tau^{\text{bar}}(B) \neq T_\tau^{\text{bar}} W_W(B).$$

Proof. Take $B = \{[0, 10)\}$, $W = [0, 1)$, and $\tau = 2$. Deletion first retains the long bar and cropping produces $[0, 1)$; cropping first produces a length-one bar which is deleted. The same phenomenon occurs for an infinite bar on a bounded window. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-009
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-028, CR-A-032
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-009; inherited at immutable v19 strength.

Theorem 3.4 (CM-v20-010: Exact crop-collapse criterion). *Let (B, W) be τ -window-adapted: every globally retained bar, including every infinite bar, is either disjoint from W or remains retained after clipping. Then, and only then,*

$$W_W T_\tau^{\text{bar}}(B) = T_\tau^{\text{bar}} W_W(B)$$

as barcode multisets including multiplicity.

Proof. Globally deleted finite bars remain deleted after clipping because clipping does not increase length. A globally retained bar contributes equally in both orders precisely under the stated all-retained-bars condition. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-010
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH2-031, CR-CH2-032, CR-A-035, CR-A-036
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-010; inherited at immutable v19 strength.

4 Metric repair and comparison-mode calculus

Theorem 4.1 (CM-v20-011: Global non-expansiveness of hard deletion fails). *There exist barcodes B, C with*

$$d_B(B, C) < d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C).$$

Proof. Fix $0 < \eta < \tau/4$ and let

$$B = \{[0, \tau + \eta)\}, \quad C = \{[0, \tau - \eta)\}.$$

Then $d_B(B, C) = 2\eta$. After deletion only the bar of B remains, so the distance is $(\tau + \eta)/2 > 2\eta$. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-011
Role / status: metric_repair_theorem / proved
Source claims: CR-CH2-010, CR-A-011
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-011; inherited at immutable v19 strength.

Theorem 4.2 (CM-v20-012: Sharp two-sided threshold-gap-safe stability). *Let B, C be finite barcodes, possibly with infinite bars. If $d_B(B, C) \leq \varepsilon$ and no finite bar of either barcode has length in the closed band*

$$[\tau - \varepsilon, \tau + \varepsilon],$$

then

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon.$$

The half-width ε is sharp for this two-sided certificate.

Proof. Choose an ε -matching. Infinite bars match only infinite bars and survive. A retained finite bar has length greater than $\tau + \varepsilon$. If its finite match were deleted, the two-sided closed-band hypothesis would force the matched length below $\tau - \varepsilon$, contradicting the fact that matched finite-bar lengths differ by at most 2ε . The argument is symmetric. Removing pairs deleted on both sides restricts the original matching to the retained barcodes with no larger cost. Sharpness follows from centered bars of lengths $\tau + c + \eta$ and $\tau - c - \eta$ for any $0 < c < \varepsilon$ and sufficiently small $\eta > 0$. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-012
Role / status: metric_repair_theorem / proved
Source claims: CR-CH2-016, CR-A-017
Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-012; inherited at immutable v19 strength.

Theorem 4.3 (CM-v20-013: Scalarized-ledger-safe threshold classification). *Let $\mathfrak{d}_{\text{met}}$ lie in the declared metric-ledger quantale, let $\varepsilon_B = \text{val}_B(\mathfrak{d}_{\text{met}})$, and suppose the record proves*

$$d_B(B, C) \leq \varepsilon_B \quad \text{and} \quad \varepsilon_B < \text{gap}_\tau(B, C).$$

Then

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon_B.$$

Proof. The strict inequality gives two-sided avoidance of the closed threshold band of half-width ε_B . Apply CM-v20-012. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-013

Role / status: conditional_core_micro_theorem / conditional

Source claims: CR-CH2-017, CR-A-018

Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified

Failure routes: FR-MISSING, FR-T3, FR-T5

Repair ancestry: CM-v19-013; inherited at immutable v19 strength.

Proposition 4.4 (CM-v20-014: Exact comparison is not a positive-radius robustness certificate). *An exact equality or profile isomorphism between two specified profiles does not imply a positive threshold-gap radius.*

Proof. A profile containing a finite bar of length exactly τ is exactly equal to itself, but has $\text{clear}_\tau = 0$. Thus exact comparison and positive-radius robustness are logically distinct. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-014

Role / status: core_micro_theorem / proved

Source claims: CR-CH1-025, CR-CH1-026, CR-CH3-009

Comparison policy: fixed_exact; **runtime modes:** exact

Failure routes: FR-MISSING, FR-T5

Repair ancestry: CM-v19-014; inherited at immutable v19 strength.

Policy 4.5 (CM-v20-015: Runtime comparison-mode trichotomy). Every consumed comparison resolves exactly one runtime mode:

$$\text{exact}, \quad \text{metric_gap_certified}, \quad \text{not_compared}.$$

The first requires an actual equality or profile isomorphism; the second requires a pre-threshold discrepancy bound and two-sided threshold-gap record; the third is admissible only when no comparison inference is consumed and cannot discharge an overlap, transport, coherence, representative, tower, or Restart obligation.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-015

Role / status: operational_policy / operational

Source claims: CR-CH1-025, CR-CH7-028, CR-CH8-010

Comparison policy: runtime_resolved; **runtime modes:** exact, metric_gap_certified, not_compared

Failure routes: FR-MISSING, FR-T3, FR-T5

Repair ancestry: CM-v19-015; inherited at immutable v19 strength.

Theorem 4.6 (CM-v20-016: Overlap transfer under a scalarized threshold-gap certificate). *For directly win-dowed pre-threshold overlap profiles $B_{i, W_{\alpha\beta}}^\alpha$ and $B_{i, W_{\alpha\beta}}^\beta$, suppose*

$$d_B(B_{i, W_{\alpha\beta}}^\alpha, B_{i, W_{\alpha\beta}}^\beta) \leq \varepsilon_{\alpha\beta, i} < \text{gap}_\tau(B_{i, W_{\alpha\beta}}^\alpha, B_{i, W_{\alpha\beta}}^\beta).$$

Then the repaired overlap profiles have bottleneck, and hence interleaving, distance at most $\varepsilon_{\alpha\beta, i}$ in the declared constructible regime.

Proof. Apply CM-v20-013 to the two directly windowed pre-threshold profiles and then the one-parameter isometry theorem. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-016
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH3-022, CR-A-041, CR-E-048
Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-016; inherited at immutable v19 strength.

5 Representative and Ext-edge micro-theorems

Theorem 5.1 (CM-v20-017: Windowed representative idempotence). *Let $C_{\tau,WF}$ be a verified window-normalized representative. If a second representative is constructed with the same window, monitored scope, and profile-equivalence policy, then*

$$C_{\tau,W}(C_{\tau,WF}) \simeq_{\text{PH}, I_{\text{mon}}} C_{\tau,WF}.$$

This is profile-level equivalence, not strict filtered equality or a global idempotent endofunctor.

Proof. Window normalization prevents further clipping, and CM-v20-003 makes threshold deletion idempotent on every monitored profile. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-017
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-B-064
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-017; inherited at immutable v19 strength.

Theorem 5.2 (CM-v20-018: Windowed representative threshold composition). *Let $\tau, \sigma > 0$. If all relevant representatives exist, are window-normalized, and use the same monitored scope and window, then*

$$C_{\sigma,W}(C_{\tau,W}F) \simeq_{\text{PH}, I_{\text{mon}}} C_{\max\{\tau,\sigma\},W}F.$$

Proof. Use CM-v20-004 on each monitored profile. Window normalization prevents a second crop from creating new short bars. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-018
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-B-065
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-018; inherited at immutable v19 strength.

Theorem 5.3 (CM-v20-019: Canonical two-term Ext^1 -edge identification). *For $A \in D^{[-1,0]}(k\text{-mod})$, there is a canonical natural isomorphism*

$$\text{Ext}^1(A, k) \cong \text{Hom}_k(H^{-1}(A), k).$$

Proof. Apply $\text{Hom}_{D^b}(-, k[1])$ to the canonical truncation triangle. Since k -vector spaces are semisimple, the outer Ext_k^1 and Ext_k^2 terms vanish, leaving the displayed natural isomorphism. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-019
Role / status: core_micro_theorem / proved
Source claims: CR-CH4-016, CR-C-011
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T2, FR-T5
Repair ancestry: CM-v19-019; inherited at immutable v19 strength.

Theorem 5.4 (CM-v20-020: Conditional one-way repaired PH_1 -to- Ext^1 bridge). *If $(F; W, \tau)$ has a complete eligible realization record and*

$$\text{Eval}_{1,W,\tau}(F) = 0,$$

then

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

Proof. Eligibility places $A = \mathcal{R}(C_{\tau,W}F)$ in $D^{[-1,0]}(k\text{-mod})$ and supplies the edge-realization isomorphism between $H^{-1}(A)$ and the repaired degree-one residual edge. Vanishing of the repaired profile gives $H^{-1}(A) = 0$; CM-v20-019 then gives the Ext vanishing. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-020
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH4-025, CR-C-026
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T2, FR-T5
Repair ancestry: CM-v19-020; inherited at immutable v19 strength.

Theorem 5.5 (CM-v20-021: Reverse Ext^1 -to-repaired-profile implication fails). *Under the standing eligibility axioms alone,*

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0 \not\Rightarrow \text{Eval}_{1,W,\tau}(F) = 0.$$

Proof. Take a nonzero repaired interval profile and the constant-zero realization with constant-zero edge extractor. The amplitude and edge-realization requirements are satisfied at the zero extracted edge, and $\text{Ext}^1(0, k) = 0$, while the repaired profile remains nonzero. Reverse recovery therefore requires a separate faithfulness theorem. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-021
Role / status: core_micro_theorem / proved
Source claims: CR-C-038, CR-CH4-022; CM-localcounterexample
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-021; inherited at immutable v19 strength.

6 Terminal-generic Type IV micro-theorems

Policy 6.1 (CM-v20-022: Actual artifact and terminal tameness are prerequisites for Type IV). Type IV is assigned exactly one status:

not_invoked, undefined, certified_zero, obstructed.

Numerical diagnostics are defined only after an actual tower comparison morphism, typed source and target, terminal-tame kernel and cokernel profiles, fixed $(W, \tau, I_{\text{mon}})$, and artifact provenance are present. Missing or ill-typed input yields undefined, never $(0, 0)$.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-022
Role / status: operational_policy / operational
Source claims: CR-CH5-003, CR-CH5-014, CR-D-017, CR-D-023
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-022; inherited at immutable v19 strength.

Theorem 6.2 (CM-v20-023: Characterization of certified Type IV-free status). *A tower diagnostic record has status `certified_zero` if and only if its comparison artifact is valid, its kernel and cokernel defect profiles are terminal-tame, and*

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0).$$

With the first two clauses valid, a nonzero pair gives `obstructed`.

Proof. These are exactly the validity, terminal-tameness, and numerical clauses defining the typed diagnostic record. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-023
Role / status: core_micro_theorem / proved
Source claims: CR-CH5-019, CR-D-026
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-023; inherited at immutable v19 strength.

Theorem 6.3 (CM-v20-024: Zero diagnostics plus no interior defects imply isomorphism). *Assume the defect profiles are terminal-tame and interval-decomposable. If the terminal-generic diagnostics vanish and the kernel and cokernel contain no interior or noncofinal interval summands, then the tower comparison artifact is an isomorphism in the declared profile category.*

Proof. The zero diagnostics exclude terminal-cofinal summands. The no-interior-defect hypothesis excludes all remaining interval summands. Thus kernel and cokernel vanish. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-024
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH5-029, CR-D-029
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-024; inherited at immutable v19 strength.

Theorem 6.4 (CM-v20-025: Full-isomorphism band implies terminal-generic stable band). *Every artifact-coherent threshold band on which each monitored tower comparison artifact is an isomorphism is a global terminal-generic Type IV stable band.*

Proof. Every isomorphism has zero kernel and cokernel, hence terminal-tame zero diagnostics and a `certified_zero` record at every threshold, assuming the remaining artifact fields are present. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-025
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH5-031, CR-D-054
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-025; inherited at immutable v19 strength.

7 Restart, obligation, and countable-audit micro-theorems

Theorem 7.1 (CM-v20-026: Finite-horizon product-weighted Restart bound). *If transitions $0 \rightarrow 1, \dots, n-1 \rightarrow n$ satisfy*

$$r_{k+1} \geq \kappa_k r_k - \eta_k, \quad 0 < \kappa_k \leq 1,$$

then

$$r_n \geq \left(\prod_{j=0}^{n-1} \kappa_j \right) r_0 - \sum_{m=0}^{n-1} \left(\prod_{j=m+1}^{n-1} \kappa_j \right) \eta_m.$$

Proof. Induct on n , substitute the previous bound into $r_{n+1} \geq \kappa_n r_n - \eta_n$, and distribute κ_n . □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-026

Role / status: core_micro_theorem / proved

Source claims: CR-CH6-032, CR-E-025

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-T3, FR-T5

Repair ancestry: CM-v19-026; inherited at immutable v19 strength.

Theorem 7.2 (CM-v20-027: Conditional countable Restart-summability principle). *Set $K_0 = 1$ and $K_n = \prod_{j=0}^{n-1} \kappa_j$. Suppose the initial reserve is positive, every transition is certified, every target record is complete with all invoked non-reserve clauses passing, no required record is undefined, and*

$$\sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < r_0.$$

Then every finite-stage reserve is positive and the ordered record family is an auditable sequential certificate chain. A uniform positive lower bound additionally requires $\inf_n K_n > 0$.

Proof. CM-v20-026 gives

$$\frac{r_n}{K_n} \geq r_0 - \sum_{m=0}^{n-1} \frac{\eta_m}{K_{m+1}} > 0.$$

The artifact and target-record hypotheses give reconstructibility. Uniform positivity does not follow if $K_n \rightarrow 0$. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-027

Role / status: conditional_core_micro_theorem / conditional

Source claims: CR-CH6-037, CR-E-028

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-T3, FR-T5

Repair ancestry: CM-v19-027; inherited at immutable v19 strength.

Theorem 7.3 (CM-v20-028: No scalar manufacture of non-scalar evidence). *No increase in a metric budget, threshold gap, reserve, or summability allowance can discharge a missing representative, actual morphism, eligibility record, terminal-tameness certificate, higher-coherence record, Claim UID, manifest field, or artifact identity.*

Proof. These obligations are typed as non-scalar atoms. The scalar ledger and the non-scalar obligation registry have disjoint semantic codomains; no scalar inequality constructs an absent typed witness. □

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-028
Role / status: core_micro_theorem / proved
Source claims: CR-CH7-015, CR-F-004
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-028; inherited at immutable v19 strength.

Theorem 7.4 (CM-v20-029: Non-compensation and dependency blocker propagation). *In the typed verdict algebra, a passing sibling cannot cancel a rejected or required undefined atom. If a required dependency is not passing, every dependent theorem-evidence obligation is blocked from pass until that dependency is repaired.*

Proof. Certificate readiness is a typed conjunction over dependency-closed obligations. Conjunction is downward closed: pass requires every required atom and dependency to pass. Therefore neither numerical surplus nor an unrelated passing atom can compensate for a blocker. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-029
Role / status: core_micro_theorem / proved
Source claims: CR-F-026, CR-F-030, CR-F-061
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-029; inherited at immutable v19 strength.

Theorem 7.5 (CM-v20-030: Finite prefixes do not certify countable readiness). *A finite prefix of a countable Restart sequence, overlap atlas, registry, or manifest cannot certify the complete countable object unless a separate theorem reduces the infinite obligation to finite data.*

Proof. Any finite prefix omits later entries and their dependencies. Without a reduction theorem, omitted required obligations remain unchecked, so complete readiness is undefined rather than passed. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-030
Role / status: core_micro_theorem / proved
Source claims: CR-F-045, CR-GAPP-012
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-030; inherited at immutable v19 strength.

8 Bridge-ready internal theorem

Theorem 8.1 (CM-v20-031: Eventually constant repaired tower with apex agreement). *Let*

$$Q_0 \rightarrow Q_1 \rightarrow Q_2 \rightarrow \cdots$$

be a repaired profile tower in a category admitting the declared colimit. Suppose there exists N such that every transition after N is an isomorphism, the apex profile Q_∞ is equipped with a declared isomorphism to the stabilized profile Q_N , and the canonical colimit-to-apex artifact is the induced comparison. Then that artifact is an isomorphism. If the Type IV input is terminal-tame and valid, its diagnostics are $(0, 0)$.

Proof. The colimit of an eventually constant diagram is canonically isomorphic to its stabilized value. Composing with the declared apex agreement gives an isomorphism. Its kernel and cokernel vanish, so the terminal-generic diagnostics vanish. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-031
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH8-012, CR-CH8-013, CR-D-047, CR-D-048
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-031; inherited at immutable v19 strength.

Remark 8.2 (Bridge boundary). CM-v20-031 is an internal bridge-ready theorem. It does not prove that any arithmetic, PDE, geometric, categorical, or cryptographic tower realizes into such a diagram. That transport is a separate Interface or Bridge Theorem.

9 Exact-invariance and terminal-generic defect calculus

Theorem 9.1 (CM-v20-032: Representative-invariance of eligible Ext^1 discharge). *Let $C_{\tau,WF}$ and $C'_{\tau,WF}$ be admissible representatives with eligible realizations*

$$A := \mathcal{R}(C_{\tau,WF}), \quad A' := \mathcal{R}(C'_{\tau,WF})$$

in $D^{[-1,0]}(k\text{-mod})$. Suppose an isomorphism $\eta : A \xrightarrow{\sim} A'$ is compatible with the declared degree-one edge-realization records. Then precomposition induces an isomorphism

$$\eta^* : \text{Ext}^1(A', k) \xrightarrow{\sim} \text{Ext}^1(A, k),$$

and consequently

$$\text{Ext}^1(A, k) = 0 \iff \text{Ext}^1(A', k) = 0.$$

Thus an eligible categorical discharge is invariant under an artifact-backed change of realization-equivalent representative.

Proof. The contravariant functor $\text{Hom}_{D(k)}(-, k[1])$ sends isomorphisms to isomorphisms. Compatibility of the edge records ensures that the two Ext groups discharge the same declared repaired degree-one clause rather than merely two unrelated derived objects. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-032
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH4-036, CR-B-057
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-032; inherited at immutable v19 strength.

Theorem 9.2 (CM-v20-033: Exact-conjugacy invariance of Type IV diagnostics). *Let $\phi : M \rightarrow N$ and $\phi' : M' \rightarrow N'$ be actual comparison morphisms and suppose there are isomorphisms $\alpha : M \xrightarrow{\sim} M'$, $\beta : N \xrightarrow{\sim} N'$ such that*

$$\beta \circ \phi = \phi' \circ \alpha.$$

Then

$$\ker \phi \cong \ker \phi', \quad \text{coker } \phi \cong \text{coker } \phi'.$$

Whenever both records are terminal-tame on the same monitored scope,

$$\mu(\phi) = \mu(\phi'), \quad \nu(\phi) = \nu(\phi'),$$

and the corresponding total collapse diagnostics agree degreewise and after aggregation.

Proof. The isomorphism α restricts to the kernels, and β descends to the cokernels. Terminal-generic dimension is invariant under persistence-module isomorphism. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-033
Role / status: core_micro_theorem / proved
Source claims: CR-D-062;CM-localproof
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-033; inherited at immutable v19 strength.

Theorem 9.3 (CM-v20-034: Terminal-generic index identity). *Let $\phi : M \rightarrow N$ be an actual morphism between terminal-tame persistence modules with finite-dimensional stabilized tails. Then*

$$\mu(\phi) - \nu(\phi) = \text{tgdim}(M) - \text{tgdim}(N).$$

Proof. Choose a common terminal parameter after which M , N , and the defect profiles are constant up to the declared tail identifications. The induced finite-dimensional map $\phi_\infty : M_\infty \rightarrow N_\infty$ satisfies

$$\dim \ker \phi_\infty - \dim \text{coker } \phi_\infty = \dim M_\infty - \dim N_\infty$$

by rank-nullity. These four dimensions are precisely the declared terminal-generic dimensions. $\square$

Corollary 9.4 (Balanced-tail consequence). *If $\text{tgdim}(M) = \text{tgdim}(N)$, then $\mu(\phi) = \nu(\phi)$. In particular, under the valid terminal-tame record, either one vanishes if and only if the other vanishes.*

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-034
Role / status: core_micro_theorem / proved
Source claims: CR-CH5-011, CR-D-013;CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-034; inherited at immutable v19 strength.

Theorem 9.5 (CM-v20-035: Direct-sum additivity of Type IV diagnostics). *For terminal-tame actual morphisms $\phi_1 : M_1 \rightarrow N_1$ and $\phi_2 : M_2 \rightarrow N_2$,*

$$\mu(\phi_1 \oplus \phi_2) = \mu(\phi_1) + \mu(\phi_2), \quad \nu(\phi_1 \oplus \phi_2) = \nu(\phi_1) + \nu(\phi_2).$$

The same statement holds for every finite direct sum and for the declared finite aggregation over monitored degrees.

Proof. Kernels and cokernels commute with finite biproducts in the persistence-module abelian category:

$$\ker(\phi_1 \oplus \phi_2) \cong \ker \phi_1 \oplus \ker \phi_2, \quad \text{coker}(\phi_1 \oplus \phi_2) \cong \text{coker } \phi_1 \oplus \text{coker } \phi_2.$$

Terminal-generic dimension is additive on finite direct sums. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-035
Role / status: core_micro_theorem / proved
Source claims: CR-CH5-015, CR-D-020;CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-035; inherited at immutable v19 strength.

Theorem 9.6 (CM-v20-036: Composition subadditivity of Type IV diagnostics). *Let*

$$M \xrightarrow{f} N \xrightarrow{g} P$$

be terminal-tame actual morphisms on a common terminal scope. Then

$$\mu(gf) \leq \mu(f) + \mu(g), \quad \nu(gf) \leq \nu(f) + \nu(g).$$

Consequently, two valid terminal-generically clean comparisons have a clean composite.

Proof. On a common stabilized tail these are finite-dimensional linear maps. The map $\ker(gf) \rightarrow \ker g$, $x \mapsto f(x)$, has kernel $\ker f$, giving the kernel inequality. Also

$$\text{rank}(gf) \geq \text{rank}(f) + \text{rank}(g) - \dim N,$$

which rearranges to the cokernel inequality. Tail dimensions equal the terminal-generic diagnostics. $\square$

Remark 9.7 (No converse). Cleanliness of gf does not force cleanliness of f or g ; defects may be killed or bypassed by composition.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-036
Role / status: core_micro_theorem / proved
Source claims: CR-D-023;CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-036; inherited at immutable v19 strength.

Theorem 9.8 (CM-v20-037: Snake-lemma control over short exact comparison diagrams). *Consider a commutative diagram of persistence modules with exact rows*

$$\begin{array}{ccccccc} 0 & \rightarrow & M' & \rightarrow & M & \rightarrow & M'' \rightarrow 0 \\ & & \downarrow \phi' & & \downarrow \phi & & \downarrow \phi'' \\ 0 & \rightarrow & N' & \rightarrow & N & \rightarrow & N'' \rightarrow 0. \end{array}$$

Assume the six modules and the vertical defect profiles are terminal-tame on a common terminal scope. Then

$$\mu(\phi) \leq \mu(\phi') + \mu(\phi''), \quad \nu(\phi) \leq \nu(\phi') + \nu(\phi'').$$

In particular, terminal-generic cleanliness of both outer vertical maps implies cleanliness of the middle map.

Proof. Pointwise exactness and the snake lemma give

$$0 \rightarrow \ker \phi' \rightarrow \ker \phi \rightarrow \ker \phi'' \rightarrow \text{coker } \phi' \rightarrow \text{coker } \phi \rightarrow \text{coker } \phi'' \rightarrow 0.$$

Pass to a common stabilized tail and take finite-dimensional vector-space dimensions. Exactness bounds the middle kernel and cokernel dimensions by the sums of their outer counterparts. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-037
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-D-023;CM-localproofviasnakelemma
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-037; inherited at immutable v19 strength.

10 Single-morphism metric bridge

Definition 10.1 (Finite-shift triviality). For $\epsilon \geq 0$, a persistence module E is ϵ -trivial if its translation morphism $E \rightarrow E(\epsilon)$ is zero.

Theorem 10.2 (CM-v20-038: Single-morphism characterization of one-parameter interleaving). *Let M, N be pointwise finite-dimensional one-parameter persistence modules and let $\delta \geq 0$. Then M and N are δ -interleaved if and only if there exists a morphism*

$$f : M \longrightarrow N(\delta)$$

whose kernel and cokernel are both 2δ -trivial.

Proof. This is Corollary 6.6 of Bauer and Lesnick (2015). Its one-parameter and pointwise finite-dimensional hypotheses are part of the theorem and are not silently weakened. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-038
Role / status: core_micro_theorem / proved
Source binding: Bauer--Lesnick2015, Corollary6.6; CM-localimportproofrecord
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: CM-v19-038; inherited at immutable v19 strength.

Theorem 10.3 (CM-v20-039: Finite-trivial defect implies terminal-generic cleanliness). *Let $f : M \rightarrow N$ be an actual morphism whose kernel and cokernel are interval-decomposable, terminal-tame, and ϵ -trivial for some finite ϵ . Then*

$$\mu(f) = \nu(f) = 0.$$

Consequently, a morphism satisfying the defect hypothesis of CM-v20-038 is terminal-generically clean whenever it is used as the declared Type IV comparison artifact on the same scope.

Proof. A terminal-cofinal interval summand has nonzero structure maps across every finite shift sufficiently far out. Such a summand cannot occur in an ϵ -trivial module. Hence neither defect profile has a terminal-cofinal summand, so both terminal-generic dimensions vanish. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-039
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-D-056; Bauer--Lesnick2015; CM-localproof
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-T5
Repair ancestry: CM-v19-039; inherited at immutable v19 strength.

Theorem 10.4 (CM-v20-040: Terminal-generic cleanliness does not imply quantitative interleaving). *For every $\delta > 0$, terminal-generic zero of an actual comparison morphism does not imply that its kernel and cokernel are 2δ -trivial, and does not imply a δ -interleaving.*

Proof. Choose $L > 2\delta$, let $M = 0$, $N = k_{[0, L)}$, and take the zero morphism $f : M \rightarrow N(\delta)$. Its kernel is zero and its cokernel is a finite interval module, so both terminal-generic diagnostics vanish. However, the cokernel is not 2δ -trivial because its bar length is greater than 2δ . Moreover

$$d_I(M, N) = d_B(\emptyset, \{[0, L)\}) = L/2 > \delta.$$

Thus Type IV cleanliness is strictly weaker than a bounded-defect interleaving certificate. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-040
Role / status: core_micro_theorem / proved
Source claims: CR-CH5-027, CR-D-038; CM-localcounterexample
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: CM-v19-040; inherited at immutable v19 strength.

11 v20 repaired-zero and detector-transport micro-theorems

Policy 11.1 (CM-v20-041: Detector stage and completeness boundary). Every detector theorem is consumed only at its registered profile stage. Nonzero soundness and zero completeness are independent. A zero detector value is theorem evidence only when the applicable endpoint, birth or germ exhaustiveness, stencil or mesh coverage, family membership, excess-source, and artifact obligations are complete; otherwise the mathematical value is undefined.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-041
Role / status: operational_policy / operational
Source claims: CR-G-003, CR-G-004, CR-CH1-006, CR-CH1-007, CR-CH1-008, CR-CH1-009
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-NONVERDICT, FR-T5
Repair ancestry: newv20detector-governancemicro-policy

Theorem 11.2 (CM-v20-042: Distance to zero of a finite repaired profile). *Let P be a finite barcode profile. If P contains an infinite bar, then $d_B(P, 0) = +\infty$. If all bars are finite, then*

$$d_B(P, 0) = \frac{1}{2} \max_{I \in P} \ell(I),$$

with $d_B(0, 0) = 0$. In particular, if P is τ -reduced and nonzero, then $d_B(P, 0) > \tau/2$.

Proof. Against the zero barcode every finite bar must be matched to the diagonal, with cost $\ell(I)/2$, while an infinite bar has no finite diagonal match. The bottleneck cost is therefore the displayed maximum. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-042
Role / status: core_micro_theorem / proved
Source claims: CR-CH2-018, CR-CH2-019, CR-A-019, CR-A-020
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 11.3 (CM-v20-043: Sharp zero separation for *au*-reduced profiles). *Let P, Q be finite τ -reduced profiles and suppose $d_B(P, Q) \leq \varepsilon$ with $2\varepsilon \leq \tau$. Then*

$$P = 0 \iff Q = 0.$$

The constant is sharp: for $2\varepsilon > \tau$ the equivalence can fail.

Proof. If $P = 0$ and $Q \neq 0$, then an infinite bar in Q forces infinite distance, while a finite retained bar has length $> \tau \geq 2\varepsilon$ and hence distance to zero $> \varepsilon$ by CM-v20-042. The converse is symmetric. For sharpness choose one bar of length L with $\tau < L \leq 2\varepsilon$. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-043

Role / status: core_micro_theorem / proved

Source claims: CR-CH2-020, CR-A-021

Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified

Failure routes: FR-MISSING, FR-T3, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 11.4 (CM-v20-044: Metric-safe repaired-zero transport). *Let B, C be finite barcode profiles and suppose $d_B(B, C) \leq \varepsilon$. Assume that no finite bar of either profile has length in $[\tau - \varepsilon, \tau + \varepsilon]$ and that $2\varepsilon \leq \tau$. Then*

$$T_\tau^{\text{bar}}(B) = 0 \iff T_\tau^{\text{bar}}(C) = 0.$$

The operational certificate uses the strict reserves $\varepsilon < \text{gap}_\tau(B, C)$ and $2\varepsilon < \tau$.

Proof. CM-v20-012 transports the bottleneck bound through hard deletion. Both outputs are τ -reduced, so CM-v20-043 transports the exact zero predicate. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-044

Role / status: conditional_core_micro_theorem / conditional

Source claims: CR-CH2-021, CR-CH2-022, CR-A-023, CR-A-024

Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified

Failure routes: FR-MISSING, FR-T3, FR-T5

Repair ancestry: new v20 micro-theorem.

Corollary 11.5 (CM-v20-045: Detector-certified metric zero transport). *Assume a complete detector certificate proves $T_\tau^{\text{bar}}(B) = 0$. If the hypotheses of CM-v20-044 hold for (B, C) , then $T_\tau^{\text{bar}}(C) = 0$. Only the certified zero predicate is transported; detector ranks, sample locations, birth sets, germ sets, and certificate size are not transported.*

Proof. Apply CM-v20-044 to the source zero statement supplied by the complete detector certificate. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-045

Role / status: conditional_core_micro_theorem / conditional

Source claims: CR-CH2-024, CR-CH2-025, CR-A-026, CR-A-027

Comparison policy: runtime_resolved; **runtime modes:** exact, metric_gap_certified, not_compared

Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T3, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 11.6 (CM-v20-046: Raw pointwise detector ranks are not bottleneck-stable). *There is no general implication*

$$d_B(B, C) \leq \varepsilon \implies \dim \text{im}(B(t) \rightarrow B(t+s)) = \dim \text{im}(C(t) \rightarrow C(t+s))$$

for fixed t, s , even for one-bar finite barcodes and arbitrarily small positive ε .

Proof. Translate one endpoint of a one-bar interval by an arbitrarily small δ across the fixed sampling point while keeping the span map nonzero on one profile and zero on the other. The bottleneck distance is at most δ , but the ranks differ. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-046

Role / status: core_micro_theorem / proved

Source claims: CR-CH2-026, CR-CH2-027, CR-A-028, CR-A-029

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-T3, FR-T5

Repair ancestry: new v20 micro-theorem.

12 Finite detector certificates and proof compression

Theorem 12.1 (CM-v20-047: Finite-stencil nonzero soundness). *Let B be a constructible windowed profile, let T be a finite subset of the effective τ -domain, and set*

$$D_{T,\tau}^{\text{stc}}(B) := \bigoplus_{t \in T} \text{im}(B(t) \rightarrow B(t + \tau)).$$

Then

$$D_{T,\tau}^{\text{stc}}(B) \neq 0 \implies T_{\tau}^{\text{bar}}(B) \neq 0.$$

No coverage hypothesis is required for this direction.

Proof. A nonzero image component is supported by an interval summand containing both t and $t + \tau$. That summand has window-relative length $> \tau$ and is retained. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-047

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-025, CR-CH3-027, CR-CH3-028, CR-B-014, CR-B-016, CR-B-017

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.2 (CM-v20-048: Birth detector zero-reflection). *For a finite left-closed right-open windowed barcode B with a certified exhaustive birth set $T_{\text{birth},\tau}(B)$,*

$$D_{T_{\text{birth},\tau}(B),\tau}^{\text{stc}}(B) = 0 \iff T_{\tau}^{\text{bar}}(B) = 0.$$

Proof. Soundness is CM-v20-047. Every retained interval $[p, q)$ satisfies $p + \tau < q$, so the structure map at its certified birth p is nonzero. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-048

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-029, CR-CH3-030, CR-B-018, CR-B-019

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.3 (CM-v20-049: Decoration-invariant germ detector zero-reflection). *For a finite decorated windowed barcode with a certified exhaustive geometric-birth set, the finite direct sum of shifted right-germ images is zero if and only if the repaired profile is zero. This statement is independent of finite-endpoint open/closed decoration.*

Proof. For an interval with geometric endpoints $p < q$, the shifted right-germ map is nonzero exactly when $p + \tau < q$. Equality gives zero because the target germ is taken strictly to the right. Summing over the exhaustive geometric-birth set yields the equivalence. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-049

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-031, CR-CH3-032, CR-CH3-033, CR-B-020, CR-B-021, CR-B-022

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.4 (CM-v20-050: Coverage-certified finite-stencil completeness). *Let T be a finite stencil. If an artifact-backed coverage certificate proves*

$$\forall I \in \text{Ret}_{\tau, W}(B), \quad T \cap A_{\tau}(I) \neq \emptyset,$$

then

$$D_{T, \tau}^{\text{stc}}(B) = 0 \iff T_{\tau}^{\text{bar}}(B) = 0.$$

Proof. CM-v20-047 gives one direction. If the repaired profile is nonzero, coverage supplies a sample in the active-start region of a retained interval, making the stencil nonzero. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-050

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-034, CR-CH3-035, CR-B-023, CR-B-024

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.5 (CM-v20-051: Family-mesh detector completeness). *Let $\mathcal{F}$ be a declared finite-barcode family on a bounded window and suppose an independently proved uniform retained-bar excess lower bound $e_{\tau, W}(\mathcal{F}) > 0$ is available. If a finite stencil has augmented mesh strictly smaller than this bound, then it is complete for every $B \in \mathcal{F}$:*

$$D_{T, \tau}^{\text{stc}}(B) = 0 \iff T_{\tau}^{\text{bar}}(B) = 0.$$

Proof. Every retained interval has an active-start interval of length at least the certified excess. The strict augmented-mesh inequality forces the stencil to meet each such interval. Apply CM-v20-050. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-051

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-036, CR-CH3-037, CR-CH3-038, CR-CH3-039, CR-B-025, CR-B-026, CR-B-027, CR-B-028

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.6 (CM-v20-052: Exact augmented-mesh stencil count). *Let the effective detector interval have length $L_{\text{eff}} \geq 0$ and let $\gamma > 0$. The minimum number of interior stencil points required to make every augmented gap at most γ is*

$$m_{\text{mesh}}(L_{\text{eff}}, \gamma) = \max \left\{ 0, \left\lceil \frac{L_{\text{eff}}}{\gamma} \right\rceil - 1 \right\}.$$

Proof. With $N = \lceil L_{\text{eff}}/\gamma \rceil$, $N - 1$ equally spaced interior points produce N gaps of size at most γ . Conversely, m points produce $m + 1$ augmented gaps whose lengths sum to L_{eff} , forcing $L_{\text{eff}} \leq (m + 1)\gamma$. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-052

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-050, CR-CH3-051, CR-B-040, CR-B-041

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-T5

Repair ancestry: new v20 micro-theorem.

Corollary 12.7 (CM-v20-053: Family-relative MinCert upper bounds). *For a detector information class Θ :*

(i) *a finite certified union of exhaustive birth sets gives $\text{MinCert}(\Theta)$ at most the size of that union;*

(ii) under CM-v20-051, every $0 < \gamma < e_{\tau, W}(\mathcal{F})$ gives

$$\text{MinCert}(\Theta) \leq m_{\text{mesh}}(L_{\text{eff}}, \gamma).$$

These are family-relative upper bounds, not retrospective objectwise compression claims.

Proof. The birth-set union is complete by CM-v20-048. The mesh bound follows from CM-v20-051 and CM-v20-052. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-053

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-047, CR-CH3-048, CR-CH3-049, CR-CH3-052, CR-B-037, CR-B-038, CR-B-039, CR-B-042

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.8 (CM-v20-054: No uniform finite detector on an unbounded birth family). *On $W = [a, +\infty)$, fix $\tau, e > 0$ and consider the translation family*

$$\mathcal{U}_{a, \tau, e} = \{[p, p + \tau + e) : p \geq a\}.$$

For every finite stencil T there exists a member whose stencil value is zero but whose repaired profile is nonzero. Consequently the corresponding family-relative MinCert is $+\infty$.

Proof. A finite stencil is bounded above. Choose p larger than every sample. The retained bar has active-start region $[p, p + e)$, disjoint from the stencil. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-054

Role / status: core_micro_theorem / proved

Source claims: CR-CH3-053, CR-B-043

Comparison policy: not_applicable; **runtime modes:** none

Failure routes: FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.9 (CM-v20-055: Exact-stage detector invariance). *An actual persistence-profile isomorphism on the declared window induces isomorphisms of all fixed-stencil image spaces and, under the same geometric-birth convention, of the right-germ detectors. Detector zero/nonzero is therefore invariant under exact source-stage isomorphism.*

Proof. Naturality gives a commutative square for each structure map; vertical isomorphisms induce isomorphisms on images. Finite direct sums and right-germ colimits preserve the resulting isomorphisms. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-055

Role / status: conditional_core_micro_theorem / conditional

Source claims: CR-CH3-043, CR-B-032, CR-B-059

Comparison policy: fixed_exact; **runtime modes:** exact

Failure routes: FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T5

Repair ancestry: new v20 micro-theorem.

Theorem 12.10 (CM-v20-056: Metric zero-verdict composition). *Suppose complete detectors zero-reflect $T_{\tau}^{\text{bar}}(B)$ and $T_{\tau}^{\text{bar}}(C)$, and the profiles satisfy the two-sided threshold-gap hypotheses with $d_B(B, C) \leq \varepsilon$ and $2\varepsilon \leq \tau$. Then the two detector values are simultaneously zero. The operational certificate uses $2\varepsilon < \tau$.*

Proof. Compose the two detector zero-reflection equivalences with CM-v20-044. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-056
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH3-044, CR-B-033, CR-B-034
Comparison policy: fixed_metric_gap_certified; **runtime modes:** metric_gap_certified
Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

13 Higher obstruction edge micro-theorems

Theorem 13.1 (CM-v20-057: Canonical higher field-coefficient Ext edge). *Let k be a field, $A \in D^b(k\text{-mod})$, and $n \geq 0$. There is a canonical isomorphism, natural contravariantly in A ,*

$$\text{Ext}^n(A, k) \cong \text{Hom}_k(H^{-n}(A), k).$$

Consequently $\text{Ext}^n(A, k) = 0$ if and only if $H^{-n}(A) = 0$.

Proof. The hyper-Ext spectral sequence has $E_2^{p,q} = \text{Ext}_k^p(H^{-q}(A), k)$. Semisimplicity of finite-dimensional vector spaces over a field kills all columns $p > 0$, leaving the stated natural edge isomorphism. Linear duality reflects zero in finite dimension. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-057
Role / status: core_micro_theorem / proved
Source claims: CR-CH4-013, CR-C-005, CR-C-006, CR-C-007, CR-C-008
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-HIGHER-EDGE, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 13.2 (CM-v20-058: Standard n -eligible higher Ext discharge). *Let $n \geq 1$ and suppose $(F; W, \tau)$ has a complete standard n -eligible realization record with edge extractor E_n and an artifact-backed identification*

$$H^{-n}(A_{n;F,W,\tau}) \cong E_n(\text{Eval}_{n,W,\tau}(F)).$$

If $E_n(0) = 0$, then

$$\text{Eval}_{n,W,\tau}(F) = 0 \implies \text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

Proof. The source zero gives $E_n(\text{Eval}) = 0$. Apply the edge-realization identification and CM-v20-057. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-058
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH4-019, CR-CH4-020, CR-C-022, CR-C-023, CR-C-024
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-HIGHER-EDGE, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 13.3 (CM-v20-059: Natural zero-reflection upgrades discharge to equivalence). *Under the hypotheses of CM-v20-058, assume the repaired profile lies in a declared class $\mathcal{P}_n$ and a separately proved natural zero-reflection package establishes*

$$E_n(P) = 0 \implies P = 0 \quad (P \in \mathcal{P}_n).$$

Then

$$\text{Eval}_{n,W,\tau}(F) = 0 \iff \text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

Proof. The forward direction is CM-v20-058. Conversely, Ext vanishing and CM-v20-057 imply $H^{-n}(A) = 0$; the edge-realization certificate gives $E_n(P) = 0$, and the independent zero-reflection theorem gives $P = 0$. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-059
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH1-012, CR-CH4-021, CR-CH4-022, CR-C-027, CR-C-028, CR-C-029, CR-C-037
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-HIGHER-EDGE, FR-DETECTOR-CERTIFICATION, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 13.4 (CM-v20-060: Synthetic duality does not create a natural bridge). *A shifted detector vector space or toy-derived object may satisfy an algebraic identity*

$$\text{Ext}^n(A^{\text{syn}}, k) \cong \text{Hom}_k(D(P), k),$$

but this does not establish a natural realization of P , representative independence, a PH_n -to- Ext^n theorem, or an external-domain bridge.

Proof. The displayed identity concerns the chosen synthetic object. The missing realization, naturality, zero-reflection, representative, and external Return arrows are independent typed obligations. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-060
Role / status: core_micro_theorem / proved
Source claims: CR-G-007, CR-CH4-039, CR-CH4-040, CR-CH4-041, CR-C-034, CR-C-035, CR-C-036
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 13.5 (CM-v20-061: Representative invariance of higher Ext discharge). *If two standard n -eligible representatives are connected by a recorded realization isomorphism compatible with their higher edge-realization certificates, then their Ext groups are identified and the truth value of the higher Ext-vanishing clause is independent of the representative choice.*

Proof. The contravariant derived Hom functor carries the realization isomorphism to an Ext isomorphism. Compatibility with the edge certificates identifies the complete discharge data. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-061
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH4-033, CR-CH4-034, CR-CH4-036, CR-B-057
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-HIGHER-EDGE, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 13.6 (CM-v20-062: Complete detector higher Ext discharge). *Let $n \geq 1$, let $(F; W, \tau)$ be standard n -eligible, and let a complete detector certificate on the exact repaired-profile stage establish that detector zero implies $\text{Eval}_{n,W,\tau}(F) = 0$. A certified zero detector value then implies*

$$\text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

If the detector is also the declared natural zero-reflecting edge, detector zero is equivalent to Ext vanishing on the registered class.

Proof. Detector completeness gives repaired-profile zero; CM-v20-058 gives Ext zero. The equivalence clause is CM-v20-059 specialized to the declared detector edge. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-062
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH4-024, CR-CH4-037, CR-C-032, CR-C-033
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-DETECTOR-NONZERO, FR-DETECTOR-CERTIFICATION, FR-HIGHER-EDGE, FR-T5
Repair ancestry: new v20 micro-theorem.

14 Terminal Type IV, transient defect, and full-isomorphism calculus

Theorem 14.1 (CM-v20-063: Terminal and interior defect separation). *For an interval-decomposable terminal-tame profile M on W , let $\text{Cof}_W(M)$ be the cofinal subprofile and $\text{Int}_W(M)$ the noncofinal subprofile. Then*

$$M \cong \text{Cof}_W(M) \oplus \text{Int}_W(M), \quad \text{tgdim}_W(M) = \#\mathcal{B}(\text{Cof}_W(M)),$$

with multiplicity. Hence $\text{tgdim}_W(M) = 0$ if and only if $M \cong \text{Int}_W(M)$.

Proof. The terminal-generic value counts exactly the terminal-cofinal interval summands. The barcode partitions canonically into cofinal and noncofinal summands. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-063
Role / status: core_micro_theorem / proved
Source claims: CR-CH5-021, CR-D-031
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 14.2 (CM-v20-064: Complete transient-defect detector characterization). *Let ϕ be a valid tower comparison morphism with constructible kernel and cokernel profiles. Rebuild complete detector certificates at source stages `kernel_defect` and `cokernel_defect`, with a declared defect threshold and either the full or interior component. Then detector zero is equivalent to zero of the corresponding hard-thresholded defect profile, and detector nonzero is equivalent to a retained defect bar.*

Proof. The interval-image, birth, germ, coverage, and mesh detector theorems apply to the supplied defect profiles after stage-specific reconstruction. No pre-threshold detector artifact is relabeled. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-064
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH5-022, CR-CH5-023, CR-CH5-024, CR-D-033, CR-D-034, CR-D-035
Comparison policy: runtime_resolved; **runtime modes:** exact, metric_gap_certified, not_compared
Failure routes: FR-MISSING, FR-TRANSIENT, FR-DETECTOR-CERTIFICATION, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 14.3 (CM-v20-065: Joint terminal and long-transient classification). *Assume a valid comparison artifact, terminal-tame defect profiles, and complete interior defect detectors. Degreewise, and after finite monitored aggregation:*

- (i) *nonzero (μ, ν) certifies terminal/cofinal Type IV, independently of any interior defect;*
- (ii) *$(\mu, \nu) = (0, 0)$ with a certified nonzero interior detector gives a long noncofinal transient defect;*

- (iii) $(\mu, \nu) = (0, 0)$ with certified-zero interior detectors excludes terminal defects and interior bars longer than the defect threshold, but not shorter defects.

Proof. CM-v20-063 classifies cofinal defects and CM-v20-064 classifies retained interior defects. Their conjunctions give the cases without compensation. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-065
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH5-026, CR-CH5-027, CR-D-037, CR-D-038
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-TRANSIENT, FR-DETECTOR-CERTIFICATION, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 14.4 (CM-v20-066: Conditional full-isomorphism upgrade). *Let ϕ be a valid morphism in the declared abelian profile category. Suppose terminal diagnostics vanish, complete interior defect detectors are certified zero at τ_{def} , and a separate short-defect exclusion certificate rules out every remaining nonzero interior bar of length at most τ_{def} . Then*

$$\ker \phi = 0, \quad \text{coker } \phi = 0,$$

so ϕ is an isomorphism.

Proof. Terminal zero excludes cofinal defect bars. Detector zero excludes interior bars longer than the defect threshold. The short-defect theorem excludes the remainder. A morphism in an abelian category with zero kernel and cokernel is an isomorphism. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-066
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH5-028, CR-CH5-029, CR-D-039, CR-D-040
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T4, FR-TRANSIENT, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 14.5 (CM-v20-067: Terminal cleanliness is strictly weaker than full or metric cleanliness). *Vanishing terminal-generic diagnostics does not imply full isomorphism, absence of finite transient defects, a quantitative interleaving bound, or finite-shift triviality of the full kernel and cokernel profiles.*

Proof. A zero morphism $0 \rightarrow k_{[0,L]}$ has zero terminal diagnostics for finite L but nonzero cokernel; choosing L arbitrarily large defeats any prescribed interleaving or finite-shift bound. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-067
Role / status: core_micro_theorem / proved
Source claims: CR-CH5-027, CR-CH5-029, CR-D-038, CR-D-040; CM-localcounterexample
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T5
Repair ancestry: new v20 micro-theorem.

15 Detector-aware Restart, atlas coverage, and finite-to-infinite reduction

Theorem 15.1 (CM-v20-068: Coverage requirement for detector atlases). *Let a global repaired-profile interface and a finite or countable family of complete local detector certificates be given. Local detector-zero*

verdicts imply global repaired-profile zero only when a separately verified coverage record proves that every global retained bar is visible in at least one local detector scope. Pairwise overlap coherence alone is insufficient.

Proof. Without coverage a retained bar may cross local boundaries or lie outside all active-start regions while every local detector is zero. With coverage, any global retained bar would create a nonzero local detector, contradicting local zero. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-068
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH6-026, CR-E-049, CR-E-050, CR-E-051, CR-E-052
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 15.2 (CM-v20-069: Detector-aware Restart handoff is nonautomatic). *A certified Restart reserve inequality does not transport a detector-zero predicate across a window transition. A valid handoff additionally requires a target-zero route and boundary-bridge coverage sufficient to expose every retained bar crossing the partition boundary.*

Proof. Restart controls normalized reserve loss, not detector completeness or bar visibility. The missing target-zero and boundary-coverage clauses are non-scalar obligations. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-069
Role / status: core_micro_theorem / proved
Source claims: CR-CH6-039, CR-CH6-040, CR-E-053, CR-E-054, CR-E-055
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-DETECTOR-CERTIFICATION, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 15.3 (CM-v20-070: Eventually constant exact finite-to-infinite reduction). *Let $M_0 \rightarrow M_1 \rightarrow \dots$ be a directed sequence with declared colimit L . Suppose all transitions are isomorphisms from some stage N onward. Then the canonical map $M_N \rightarrow L$ is an isomorphism. If a compatible apex cocone has $M_N \rightarrow A_\infty$ an isomorphism, the induced comparison $L \rightarrow A_\infty$ is an isomorphism; in an abelian profile category its kernel and cokernel vanish, all terminal diagnostics vanish, and every complete full or interior defect detector has zero value.*

Proof. The stabilized value supplies a cocone inverse to the colimit map by the universal property. Apex agreement factors as the composite of two isomorphisms. The defect conclusions follow from zero kernel and cokernel. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-070
Role / status: conditional_core_micro_theorem / conditional
Source claims: CR-CH8-012, CR-CH8-013, CR-D-046, CR-D-047, CR-D-048
Comparison policy: fixed_exact; **runtime modes:** exact
Failure routes: FR-MISSING, FR-T3, FR-T4, FR-TRANSIENT, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 15.4 (CM-v20-071: Finite prefixes do not determine an infinite reduction). *No finite prefix of a directed sequence determines its colimit, eventual constancy, apex agreement, or finite-to-infinite comparison defect without a separate reduction theorem and complete tail evidence.*

Proof. Two systems may agree through any prescribed finite stage, while one remains zero and the other becomes a nonzero constant system at the next stage. Their colimits differ. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-071
Role / status: core_micro_theorem / proved
Source claims: CR-CH8-020, CR-F-045, CR-GAPP-012
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-T3, FR-T5
Repair ancestry: new v20 micro-theorem.

Theorem 15.5 (CM-v20-072: Layered non-compensation for v20 certificates). *A passing detector layer cannot compensate for missing higher-edge eligibility; a passing higher-edge layer cannot compensate for an incomplete finite-to-infinite reduction; clean terminal diagnostics cannot compensate for transient-defect uncertainty; and artifact closure cannot compensate for a false theorem hypothesis. Every required layer and dependency must pass independently.*

Proof. The v20 readiness object is a dependency-closed typed conjunction over detector, profile, derived, tower/defect, reduction/Return, and governance layers. Conjunction admits no cancellation across semantic sorts. $\square$

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-072
Role / status: core_micro_theorem / proved
Source claims: CR-F-030, CR-F-060, CR-F-061, CR-F-069, CR-GAPP-048
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry: new v20 micro-theorem.

16 CM register closure and migration

Policy 16.1 (CM-v20-073: CM-v20 register closure and source-binding discipline). A CM-v20 package is register-closed only when every CM UID is unique and stable, every source Claim UID or primary-source locator resolves, every conditional hypothesis is displayed, the comparison policy is compatible with its runtime mode, detector stage and completeness sources are explicit, failure routes are typed, human-readable and machine-readable registers agree, and no deprecated, synthetic, operational, or non-claim item is consumed as theorem evidence. Final promotion into CR-v20 requires a separate immutable promotion receipt.

CM UID: ak-hpdst:20.0.0:CM-v20:CM-v20-073
Role / status: operational_policy / operational
Source claims: CR-G-002, CR-G-014, CR-G-015, CR-G-016, CR-G-017, CR-F-070, CR-GAPP-038, CR-GAPP-039
Comparison policy: not_applicable; **runtime modes:** none
Failure routes: FR-MISSING, FR-NONVERDICT, FR-T5
Repair ancestry: supersedes CM-v19-041 closure record.

16.1 Canonical CM-v20 register summary

CM ID	Title	Status	Comparison policy
CM-v20-001	Core micro-theorem boundary and no external upgrade	operational	not_applicable
CM-v20-002	Infinite-bar threshold invariance	proved	not_applicable

CM ID	Title	Status	Comparison policy
CM-v20-003	Hard threshold deletion idempotence	proved	not_applicable
CM-v20-004	Hard threshold composition	proved	not_applicable
CM-v20-005	Retained-profile threshold monotonicity	proved	not_applicable
CM-v20-006	Positive-threshold short bars are not Serre	proved	not_applicable
CM-v20-007	Nested window restriction identity	proved	not_applicable
CM-v20-008	Fixed window-first repaired evaluation	operational	not_applicable
CM-v20-009	No unconditional crop-collapse commutation	proved	not_applicable
CM-v20-010	Exact crop-collapse criterion	conditional	fixed_exact
CM-v20-011	Global non-expansiveness of hard deletion fails	proved	not_applicable
CM-v20-012	Sharp two-sided threshold-gap-safe stability	proved	fixed_metric_gap _certified
CM-v20-013	Scalarized-ledger-safe threshold classification	conditional	fixed_metric_gap _certified
CM-v20-014	Exact comparison is not a positive-radius robustness certificate	proved	fixed_exact
CM-v20-015	Runtime comparison-mode trichotomy	operational	runtime_resolved
CM-v20-016	Overlap transfer under a scalarized threshold-gap certificate	conditional	fixed_metric_gap _certified
CM-v20-017	Windowed representative idempotence	conditional	fixed_exact
CM-v20-018	Windowed representative threshold composition	conditional	fixed_exact
CM-v20-019	Canonical two-term Ext^1 -edge identification	proved	not_applicable
CM-v20-020	Conditional one-way repaired PH_1 -to- Ext^1 bridge	conditional	not_applicable
CM-v20-021	Reverse Ext^1 -to-repaired-profile implication fails	proved	not_applicable
CM-v20-022	Actual artifact and terminal tameness are prerequisites for Type IV	operational	not_applicable
CM-v20-023	Characterization of certified Type IV-free status	proved	not_applicable
CM-v20-024	Zero diagnostics plus no interior defects imply isomorphism	conditional	not_applicable
CM-v20-025	Full-isomorphism band implies terminal-generic stable band	conditional	not_applicable
CM-v20-026	Finite-horizon product-weighted Restart bound	proved	not_applicable
CM-v20-027	Conditional countable Restart-summability principle	conditional	not_applicable
CM-v20-028	No scalar manufacture of non-scalar evidence	proved	not_applicable
CM-v20-029	Non-compensation and dependency blocker propagation	proved	not_applicable
CM-v20-030	Finite prefixes do not certify countable readiness	proved	not_applicable
CM-v20-031	Eventually constant repaired tower with apex agreement	conditional	fixed_exact
CM-v20-032	Representative-invariance of eligible Ext^1 discharge	conditional	fixed_exact
CM-v20-033	Exact-conjugacy invariance of Type IV diagnostics	proved	fixed_exact
CM-v20-034	Terminal-generic index identity	proved	not_applicable
CM-v20-035	Direct-sum additivity of Type IV diagnostics	proved	not_applicable
CM-v20-036	Composition subadditivity of Type IV diagnostics	proved	not_applicable
CM-v20-037	Snake-lemma control over short exact comparison diagrams	conditional	not_applicable
CM-v20-038	Single-morphism characterization of one-parameter interleaving	proved	not_applicable
CM-v20-039	Finite-trivial defect implies terminal-generic cleanliness	conditional	not_applicable
CM-v20-040	Terminal-generic cleanliness does not imply quantitative interleaving	proved	not_applicable
CM-v20-041	Detector stage and completeness boundary	operational	not_applicable
CM-v20-042	Distance to zero of a finite repaired profile	proved	not_applicable
CM-v20-043	Sharp zero separation for au -reduced profiles	proved	fixed_metric_gap _certified
CM-v20-044	Metric-safe repaired-zero transport	conditional	fixed_metric_gap _certified

CM ID	Title	Status	Comparison policy
CM-v20-045	Detector-certified metric zero transport	conditional	runtime_resolved
CM-v20-046	Raw pointwise detector ranks are not bottleneck-stable	proved	not_applicable
CM-v20-047	Finite-stencil nonzero soundness	proved	not_applicable
CM-v20-048	Birth detector zero-reflection	proved	not_applicable
CM-v20-049	Decoration-invariant germ detector zero-reflection	proved	not_applicable
CM-v20-050	Coverage-certified finite-stencil completeness	proved	not_applicable
CM-v20-051	Family-mesh detector completeness	proved	not_applicable
CM-v20-052	Exact augmented-mesh stencil count	proved	not_applicable
CM-v20-053	Family-relative MinCert upper bounds	proved	not_applicable
CM-v20-054	No uniform finite detector on an unbounded birth family	proved	not_applicable
CM-v20-055	Detector invariance under exact source-stage isomorphism	conditional	fixed_exact
CM-v20-056	Metric zero-verdict composition	conditional	fixed_metric_gap_certified
CM-v20-057	Canonical higher field-coefficient Ext edge	proved	not_applicable
CM-v20-058	Standard n -eligible higher Ext discharge	conditional	not_applicable
CM-v20-059	Natural zero-reflection upgrades discharge to equivalence	conditional	not_applicable
CM-v20-060	Synthetic duality does not create a natural bridge	proved	not_applicable
CM-v20-061	Representative invariance of higher Ext discharge	conditional	fixed_exact
CM-v20-062	Complete detector higher Ext discharge	conditional	not_applicable
CM-v20-063	Terminal and interior defect separation	proved	not_applicable
CM-v20-064	Complete transient-defect detector characterization	conditional	runtime_resolved
CM-v20-065	Joint terminal and long-transient classification	conditional	not_applicable
CM-v20-066	Conditional full-isomorphism upgrade	conditional	not_applicable
CM-v20-067	Terminal cleanliness is strictly weaker than full or metric cleanliness	proved	not_applicable
CM-v20-068	Detector-atlas zero requires global retained-bar coverage	conditional	not_applicable
CM-v20-069	Detector-aware Restart handoff is nonautomatic	proved	not_applicable
CM-v20-070	Eventually constant exact finite-to-infinite reduction	conditional	fixed_exact
CM-v20-071	Finite prefixes do not determine an infinite reduction	proved	not_applicable
CM-v20-072	Layered non-compensation for v20 certificates	proved	not_applicable
CM-v20-073	CM-v20 register closure and source-binding discipline	operational	not_applicable

16.2 Migration rule

The canonical UIDs CM-v20-001–CM-v20-040 conservatively succeed the same-numbered CM-v19 items at unchanged or weaker-safe mathematical strength. The former closure item CM-v19-041 is superseded by CM-v20-073. Item CM-v20-041 and items CM-v20-042–CM-v20-072 are new v20 records. No bare legacy identifier resolves without its theory-version namespace, and no v20 registration mutates a v19 source artifact or verdict.

17 Explicit exclusions

- (i) No global non-expansiveness, exactness, Serre localization, or strict global filtered functor for hard positive-threshold deletion.
- (ii) No metric transport through hard deletion without the two-sided threshold-gap hypotheses and the actual discrepancy bound.

- (iii) No repaired-zero transport without post-threshold zero separation; raw pointwise detector rank is not bottleneck-stable.
- (iv) No detector-zero conclusion from soundness alone, from a finite observation alone, or from an unverified birth, germ, stencil, mesh, family, endpoint, stage, or coverage record.
- (v) No family-relative MinCert claim reconstructed retrospectively from a single already-known target without the declared information class and noncircular evidence source.
- (vi) No uniform complete finite detector on a right-unbounded unbounded-birth family without additional stabilization, finite-birth, global-bound, cofinal, or finite-to-infinite structure.
- (vii) No higher PH-to-Ext equivalence without a complete standard n -eligible record and a separately proved natural zero-reflection theorem.
- (viii) No synthetic or toy shifted duality promoted to a natural realization theorem, representative-independent bridge, or external-domain theorem.
- (ix) No terminal Type IV zero interpreted as full defect vanishing, transient cleanliness, full isomorphism, or quantitative interleaving.
- (x) No defect detector reused across profile stages without reconstructing the actual kernel or cokernel profile and a new stage-specific certificate.
- (xi) No source/apex metric bounds automatically transported to kernel or cokernel defect profiles.
- (xii) No Restart reserve inequality interpreted as detector-zero transport, boundary coverage, higher coherence, or finite-to-infinite reduction.
- (xiii) No finite prefix interpreted as a complete countable atlas, tail theorem, eventual-constancy package, colimit certificate, or external Return Theorem.
- (xiv) No numerical slack manufactures a representative, morphism, eligibility record, naturality family, completeness proof, terminal-tameness proof, Claim UID, schema field, or artifact identity.
- (xv) No registration, successful build, package closure, replay, Search output, AI confidence, or human approval is mathematical proof by itself.
- (xvi) No normative Latin u for the Type IV cokernel diagnostic; v is canonical.

18 Primary literature and source regime

The imported single-morphism interleaving characterization is used only in the pointwise finite-dimensional one-parameter regime and is bound to U. Bauer and M. Lesnick, *Induced Matchings and the Algebraic Stability of Persistence Barcodes*, Journal of Computational Geometry 6(2) (2015), Corollary 6.6. All other v20 items are proved internally or extracted from the immutable Main and Foundation source claims listed in their metadata.

19 Provenance and status

This document was reconstructed from the submitted CM-v19 TeX source and audited against the canonical v20 Main, corrected Foundation A–G, and CR-v20 baseline artifacts. The inherited metric, crop, representative, Type IV, Restart, obligation, and single-morphism items retain their exact v19 guard-rails. The new v20 contribution is the reusable finite-certificate, higher-edge, transient-defect, detector-atlas, and finite-to-infinite reduction layer. CM-v20 remains internal Core theorem evidence only at the strength of each displayed statement and metadata record. CR-v20 promotion and TB-v20 consumption are subsequent governance actions.